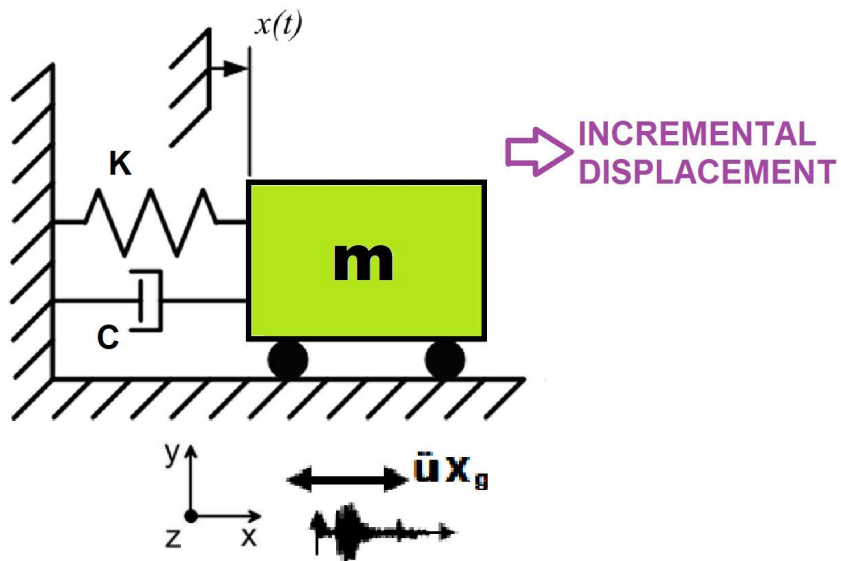


>> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST MERCIFUL <<

DETERMINATION OF OPTIMUM STRUCTURAL DAMPER STIFFNESS FROM ENERGY DISSIPATION CAPACITY INDEX USING FINITE-DIFFERENCE NEWTON ITERATION AND OPENSEES

THIS PROGRAM WRITTEN BY SALAR DELAVAR GHASHGHAEI (QASHQAI)



$$\text{Structural Ductility Damage Index} = \frac{\Delta_d - \Delta_y}{\Delta_u - \Delta_y}$$

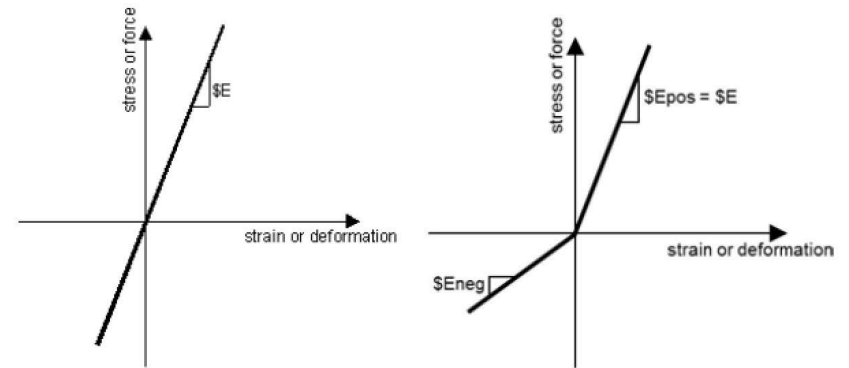
Δ_d = Lateral Displaement from Dynamic Analysis

Δ_y = Lateral Yield Displaement from Pushover Analysis

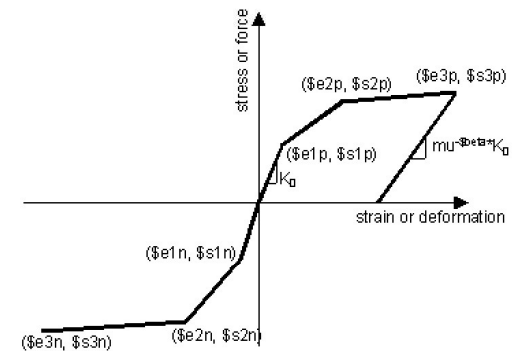
Δ_u = Lateral Ultimate Displaement from Pushover Analysis

$$P(t) = P_0 e^{-0.05\bar{\omega}t} \sin(\bar{\omega}t)$$

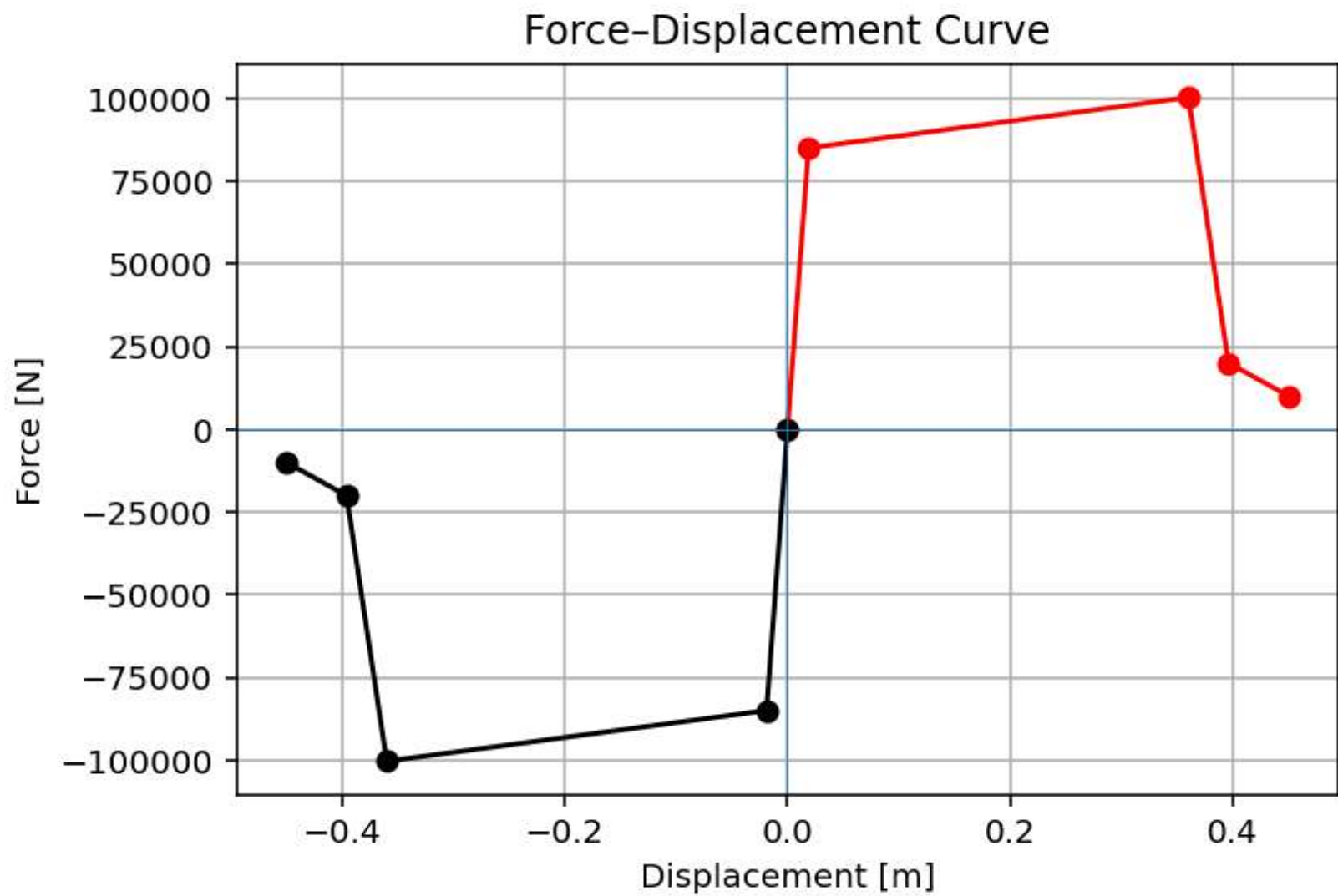
$$m\ddot{u} + c\dot{u} + ku = P(t)$$



ELASTIC MATERIAL



INELASTIC MATERIAL



1#####
2# >> IN THE NAME OF ALLAH, THE MOST GRACIOUS, THE MOST ME
3# DETERMINATION OF OPTIMUM STRUCTURAL DAMPER STIFFNESS FROM ENERGY DISSIP
4# FINITE-DIFFERENCE NEWTON ITERATION AND OPEN
5#-----
6# COMPREHENSIVE NONLINEAR SEISMIC ASSESSMENT OF A SINGLE-DEGREE-FREEDOM STR
7# FOR STATIC PUSHOVER, CYCLIC DEGRADATION, STATIC TIME-HISTORY AND DYNAM
8#-----
9# P(t) = P0 sin(wt)
10# P(t) = P0 exp(-0.05wt) sin(wt)
11#-----
12# ASSESSMENT OF DUCTILITY DAMAGE INDICES FOR STRUCTURAL ELEMENTS AN
13# OF ENERGY DISSIPATION CAPACITY IND
14#-----
15# THIS PYTHON SCRIPT WRITTEN BY SALAR DELAVAR GHASHGHAEI
16# EMAIL: salar.d.ghashghaei@gmail.com
17#####
18###
19Nonlinear Seismic Performance Assessment of a SDOF:
20An OpenSeesPy Framework for Material and Geometric Nonlinearity Under Stati
21
22This OpenSeesPy script performs rigorous nonlinear static and dynamic analy
23for performance-based earthquake engineering. The 2D model incorporates bo
24(elastic-perfectly plastic or hysteretic steel with strain hardening)
25and geometric nonlinearity (corotational truss formulation) to capture P-d
26and large displacements-essential for collapse assessment.
27
28Eight analysis protocols are implemented:
29(1) [PERIOD] : Structural Period
30(2) [STATIC] : Gravity load analysis establishing dead load state
31(3) [PUSHOVER] : Displacement-controlled pushover generating full capacity
32and plastic mechanism identification
33(4) [CYCLIC_DISPLACEMENT] : Symmetric cyclic displacement protocol capturin
34pinching behavior, and energy dissipation degradation

Cyclic Loading - Dissipated Energy (Convex Hull)

The main plot displays the relationship between Base Shear (N) on the y-axis (ranging from -100,000 to 100,000) and Displacement (m) on the x-axis (ranging from -0.3 to 0.3). It features a black hysteresis curve, a red dashed convex hull, and a red shaded area representing the energy dissipation. The plot is titled 'Cyclic Loading - Dissipated Energy (Convex Hull)'.

Legend:

- Hysteresis Curve
- Convex Hull
- Hull Area (Energy)

Continuous Response - Dissipated Energy (Convex Hull)

This plot shows the continuous response of the system, including the hysteresis curve, convex hull, and hull area (energy). The y-axis represents Base Shear (N) and the x-axis represents Displacement (m).

Cyclic Loading - Dissipated Energy (Convex Hull)

This plot shows the cyclic loading response, including the hysteresis curve, convex hull, and hull area (energy). The y-axis represents Base Shear (N) and the x-axis represents Displacement (m).

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\ DELL\Desktop\OPENSEES_FILES\+EIGHT_ANALY...TION_DECI_SDOF\K_Damper_OPTIMIZATION_DECI_SDOF.py

K_Damper_OPTIMIZATION_DECI_SDOF.py DISSIPATED_ENERGY_WITH_PLOT_FUN.py

52 %

IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 51%

OPTIMIZATION PROBLEM FOR ENERGY DISSIPATION CAPACITY INDEX (EDCI)

Spyder (Python 3.12)

File Edit Search Source Run Debug Consoles Projects Tools View Help

C:\Users\ DELL\Desktop\OPENSEES_FILES\+HEIGHT_ANALY...TION_DECI_SDOF\K_Damper_OPTIMIZATION_DECI_SDOF.py

K_Damper_OPTIMIZATION_DECI_SDOF.py DISSIPATED_ENERGY_WITH_PLOT_FUN.py

```
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```

Console 1/A X

```
+-----+
| lambda | omega | period | frequency |
+-----+
| 7.609e+02 | 27.5837 | 0.2278 | 4.3901 |
+-----+

Time: 20.0000, Displacement: 0.0353 mm, Reaction: -20.00 N

SEISMIC ANALYSIS DONE.

Xmax: 352983904.84063834
Dissipated Energy from Earthquake= 17677.39 N·m
Dissipated Energy from Cyclic Displacement= 88386.96 N·m

DISSIPATED ENERGY CAPACITY INDEX: 20.000 [%]

ZONE 2: MINOR DAMAGE

Fmax: -2.0879298290310544e-11
DF: -2.0877521933471144e-08
DX: -0.0
IT: 11 - RESIDUAL: 0.0 X: 352983904.83963835

Optimum X (DAMPER STIFFNESS): 352983904.8396
Iteration Counts: 11
Convergence Residual: 0.000000000e+00

Total time (s): 92.8125
```

IPython Console Files Help Variable Explorer Debugger Plots History

Inline Conda: anaconda3 (Python 3.12.7) ✓ LSP: Python Line 1, Col 1 UTF-8 CRLF RW Mem 51%

Dissipated Energy from Earthquake= 17677.39 N·m

Dissipated Energy from Cyclic Displacement= 88386.96 N·m

DISSIPATED ENERGY CAPACITY INDEX: 20.000 [%]

ZONE 2: MINOR DAMAGE

Fmax: -2.0879298290310544e-11

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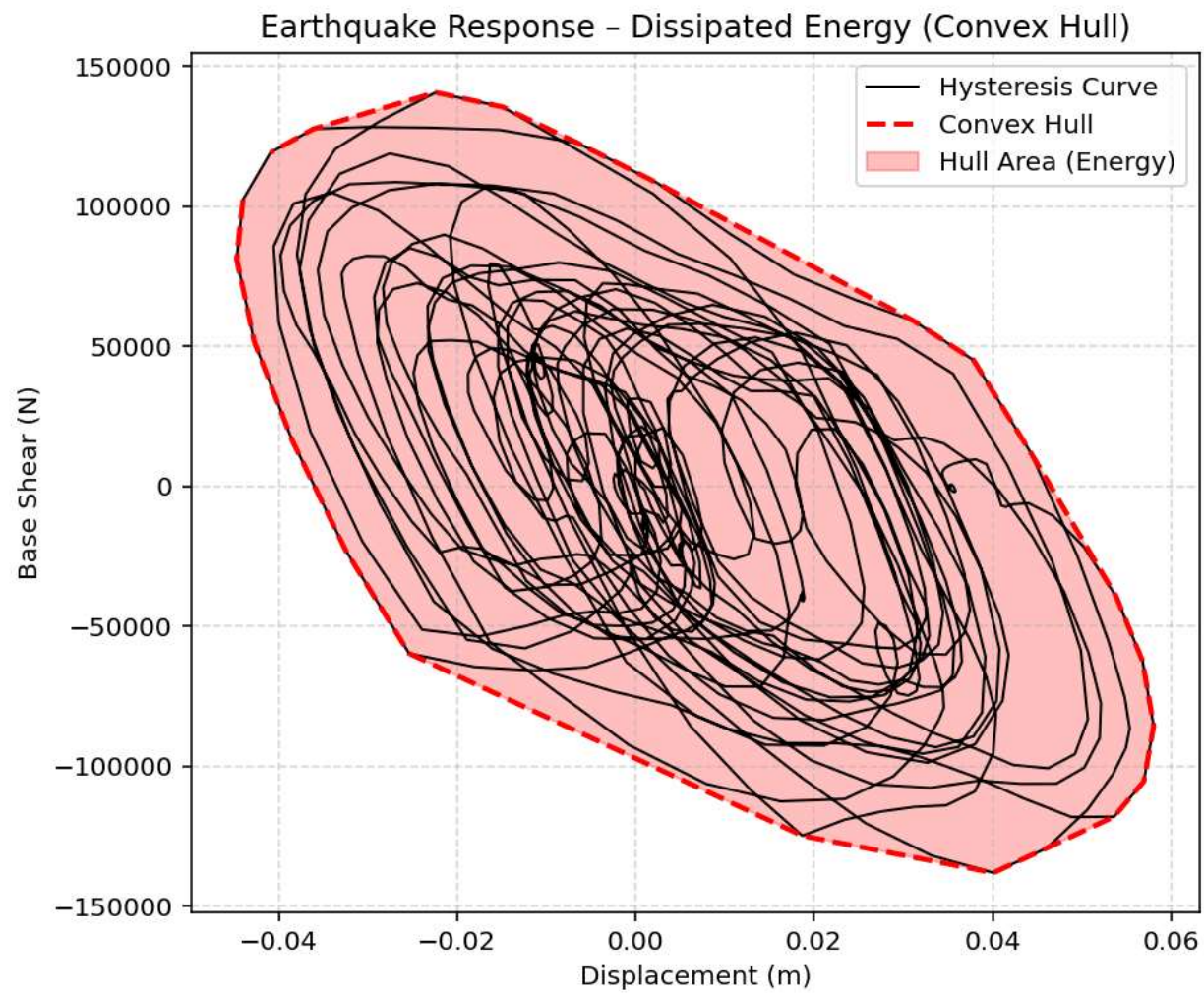
DX: -0.0

IT: 11 - RESIDUAL: 0.0 X: 352983904.83963835

Optimum X (DAMPER STIFFNESS): 352983904.8396

Iteration Counts: 11

Convergence Residual: 0.0000000000e+00



Cyclic Loading - Dissipated Energy (Convex Hull)

